Labsheet 1 : BTCS301P-AS01

Computer Vision Lab

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In [ ]:
```python
# Khushi Choudhary CU240251564
import cv2
import numpy as np
import matplotlib.pyplot as plt
# Load the image (OpenCV reads in BGR format by default)
bgr_img = cv2.imread('sample_img1.jpg')
if bgr_img is None:
    raise FileNotFoundError("Image not found. Ensure 'sample.jpg' exists in the

# Convert BGR to RGB for correct display with Matplotlib
rgb_img = cv2.cvtColor(bgr_img, cv2.COLOR_BGR2RGB)
# Display properties
h, w, c = rgb_img.shape
print(f"Height (Rows)     : {h} px")
print(f"Width (Columns)   : {w} px")
print(f"Channels          : {c}")
print(f"Data Type         : {rgb_img.dtype}")
print(f"Min/Max Intensity : {rgb_img.min()} / {rgb_img.max()}")

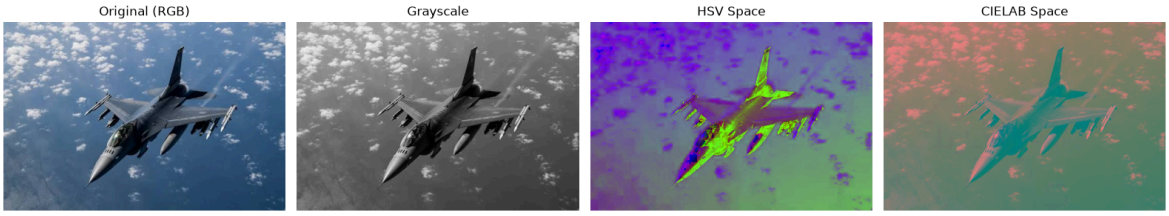
# Show image
plt.figure(figsize=(6, 4))
plt.imshow(rgb_img)
plt.title("Original Image (RGB)")
plt.axis("off")
plt.show()
```

```
Height (Rows)     : 365 px
Width (Columns)   : 547 px
Channels          : 3
Data Type         : uint8
Min/Max Intensity : 0 / 255
```



In [ ]:
```
# Save in different formats
cv2.imwrite('output_compressed.jpg', bgr_img, [cv2.IMWRITE_JPEG_QUALITY, 50])
cv2.imwrite('output_lossless.png', bgr_img)
# Color Space Conversions
gray_img = cv2.cvtColor(bgr_img, cv2.COLOR_BGR2GRAY)
hsv_img  = cv2.cvtColor(bgr_img, cv2.COLOR_BGR2HSV)
lab_img  = cv2.cvtColor(bgr_img, cv2.COLOR_BGR2LAB)

# Visualization
fig, axes = plt.subplots(1, 4, figsize=(16, 4))
axes[0].imshow(rgb_img);          axes[0].set_title("Original (RGB)")
axes[1].imshow(gray_img, cmap='gray'); axes[1].set_title("Grayscale")
axes[2].imshow(hsv_img);          axes[2].set_title("HSV Space")
axes[3].imshow(lab_img);          axes[3].set_title("CIELAB Space")
for ax in axes: ax.axis("off")
plt.tight_layout()
plt.show()
```



In [ ]:
```
# 1. Resizing (Half size and double size)
resized_half = cv2.resize(rgb_img, (w // 2, h // 2), interpolation=cv2.INTER_ARE
# 2. Flipping
flipped_h = cv2.flip(rgb_img, 1)  # 1 = Horizontal flip
# 3. Rotation (45 degrees about center)
center = (w // 2, h // 2)
rot_matrix = cv2.getRotationMatrix2D(center, angle=45, scale=1.0)
rotated = cv2.warpAffine(rgb_img, rot_matrix, (w, h))
# 4. Image Negative (Inversion)
img_negative = 255 - rgb_img
# 5. Region of Interest (ROI) - Center Crop
ymin, ymax = int(h * 0.25), int(h * 0.75)
xmin, xmax = int(w * 0.25), int(w * 0.75)
```

```
roi = rgb_img[ymin:ymax, xmin:xmax]

# Visualization grid
fig, axes = plt.subplots(2, 3, figsize=(14, 8))
axes[0, 0].imshow(resized_half);  axes[0, 0].set_title(f"Resized Half ({w//2}x{h
axes[0, 1].imshow(flipped_h);     axes[0, 1].set_title("Horizontal Flip")
axes[0, 2].imshow(rotated);       axes[0, 2].set_title("Rotated 45°")
axes[1, 0].imshow(img_negative);  axes[1, 0].set_title("Negative (Inverted)")
axes[1, 1].imshow(roi);           axes[1, 1].set_title("Cropped ROI")
axes[1, 2].axis("off")            # Empty placeholder cell

for ax in axes.flat: ax.axis("off")
plt.tight_layout()
plt.show()
```



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```
# Khushi Choudhary CU240251564
import cv2
import numpy as np
import matplotlib.pyplot as plt
# Load the image (OpenCV reads in BGR format by default)
bgr_img = cv2.imread('sample_img2.jpg')
if bgr_img is None:
    raise FileNotFoundError("Image not found. Ensure 'sample.jpg' exists in the

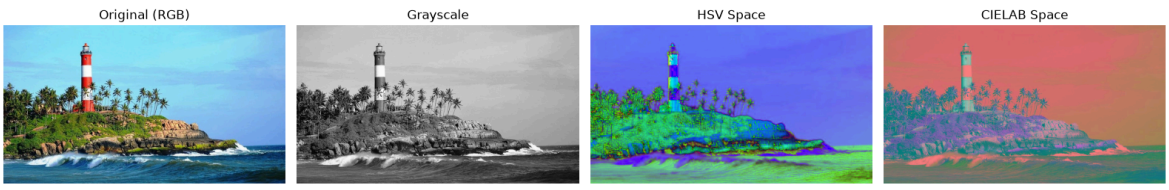
# Convert BGR to RGB for correct display with Matplotlib
rgb_img = cv2.cvtColor(bgr_img, cv2.COLOR_BGR2RGB)
# Display properties
h, w, c = rgb_img.shape
print(f"Height (Rows)     : {h} px")
print(f"Width (Columns)   : {w} px")
print(f"Channels          : {c}")
print(f"Data Type         : {rgb_img.dtype}")
print(f"Min/Max Intensity : {rgb_img.min()} / {rgb_img.max()}")

# Show image
plt.figure(figsize=(6, 4))
plt.imshow(rgb_img)
plt.title("Original Image (RGB)")
plt.axis("off")
plt.show()
```

```
Height (Rows)     : 500 px
Width (Columns)   : 900 px
Channels          : 3
Data Type         : uint8
Min/Max Intensity : 0 / 255
```



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# Save in different formats
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cv2.imwrite('output_lossless.png', bgr_img)
# Color Space Conversions
gray_img = cv2.cvtColor(bgr_img, cv2.COLOR_BGR2GRAY)
hsv_img  = cv2.cvtColor(bgr_img, cv2.COLOR_BGR2HSV)
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axes[2].imshow(hsv_img);          axes[2].set_title("HSV Space")
axes[3].imshow(lab_img);          axes[3].set_title("CIELAB Space")
for ax in axes: ax.axis("off")
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plt.show()
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center = (w // 2, h // 2)
rot_matrix = cv2.getRotationMatrix2D(center, angle=45, scale=1.0)
rotated = cv2.warpAffine(rgb_img, rot_matrix, (w, h))
# 4. Image Negative (Inversion)
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axes[0, 1].imshow(flipped_h);     axes[0, 1].set_title("Horizontal Flip")
axes[0, 2].imshow(rotated);       axes[0, 2].set_title("Rotated 45°")
axes[1, 0].imshow(img_negative);  axes[1, 0].set_title("Negative (Inverted)")
axes[1, 1].imshow(roi);           axes[1, 1].set_title("Cropped ROI")
axes[1, 2].axis("off")

for ax in axes.flat: ax.axis("off")
plt.tight_layout()
plt.show()
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